

Stochastic Optimization of Airbag Deployment Parameters

ME 701 Course Project

Jaladhi Joshi (22B0994) Manav Agrawal (22B1253)

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Problem Statement

- Goal: choose airbag deployment parameters that reduce injury risk under uncertain crash conditions.
- We optimize a fixed airbag deployment algorithm before the crash, then test it on many random crash scenarios.

Algorithm variables

- trigger delay (ms)
- early-stage pressure (kPa)
- mid-stage pressure (kPa)
- late-stage pressure (kPa)
- vent rate

Main uncertainties

- crash speed and pulse duration
- impact angle
- occupant seat offset
- belt usage
- sensing / trigger bias and delay

Stochastic algorithm-optimization problem

$$\min_{x \in \mathcal{X}} \left(w_1 \mathbb{E}[\text{HIC}] + w_2 \mathbb{E}[D_{\text{chest}}] + w_3 \mathbb{E}[E_{\text{deploy}}] + w_4 \text{CVaR}_{90}(\text{HIC}) \right)$$

where $x = [d, p_e, p_m, p_l, v]$ is the deployment algorithm.

- $\mathbb{E}[\text{HIC}]$: average head injury risk over stochastic scenarios.
- $\mathbb{E}[D_{\text{chest}}]$: average chest deflection proxy.
- $\mathbb{E}[E_{\text{deploy}}]$: average deployment energy used by the inflator / vent system.
- $\text{CVaR}_{90}(\text{HIC})$: tail-risk term that penalizes the worst 10% HIC outcomes.
- Additional penalties are added when:
 - failure rate for HIC exceeds the desired limit
 - false deployment rate becomes too high
 - average chest deflection becomes too high

How Crash Scenarios Are Sampled

Scenario generation idea

Each crash realization is sampled from a joint probability distribution over uncertain parameters,

$$\omega = (v_{\text{crash}}, T_{\text{pulse}}, \theta, s_{\text{seat}}, b, \delta_{\text{sensor}}, \delta_{\text{trigger}}) \sim P_{\text{default}} \text{ or } P_{\text{shift}}$$

and the simulator evaluates one loss value $J(x; \omega)$ for the chosen algorithm x .

$$\delta_{\text{trigger}} \sim \mathcal{N}(0, \sigma_{\text{trigger}}^2), \quad T_{\text{pulse}} \sim \text{clipped Normal}, \quad \delta_{\text{sensor}} \sim \text{clipped OU noise}$$

$$\mathbb{E}[J(x)] \approx \frac{1}{N} \sum_{i=1}^N J(x; \omega_i), \quad \omega_i \sim P_{\text{default}} \text{ or } P_{\text{shift}}$$

Evaluation Profiles

Default profile

- baseline training / evaluation distribution
- moderate crash severity mix
- higher belt-use probability
- smaller occupant-offset bias

Shifted profile

- harsher out-of-distribution test
- higher speeds and more obliquity
- lower belt-use probability
- larger occupant-offset bias
- noisier sensing and shorter pulses

Why this matters

An algorithm that looks good on the default distribution may degrade when crash statistics shift.

Methodology Pipeline

- ① Generate many crash realizations representing uncertainty in impact severity, occupant state, and sensing.
- ② For each candidate algorithm, simulate crash dynamics over the full impact event.
- ③ Convert the resulting traces into injury and efficiency metrics such as HIC, chest deflection, and energy.
- ④ Aggregate those scenario-level outcomes into a stochastic objective.
- ⑤ Use optimization or policy evaluation to obtain one algorithm from each baseline or proposed approach.
- ⑥ Evaluate every algorithm on common iid and shifted test sets for a fair comparison.

Training and Evaluation Setup

- Training scenarios:
 - 500 default scenarios for robust training
 - 300 shifted scenarios for shift-aware training
- Evaluation scenarios:
 - 3000 iid scenarios
 - 3000 shifted scenarios
- Optimization strategy:
 - global search for broad exploration
 - local refinement for final tuning
 - additional shift-focused local refinement for the final algorithm

How We Simulated the System

- Reduced-order 1D forward occupant model.
- Separate torso and head states are simulated over time.
- Crash pulse drives the occupant forward.
- Seatbelt model includes slack, damping, load limiting, and pretensioner action.
- Airbag model includes delayed triggering, staged target pressure, inflation lag, and vent losses.
- Contact terms model torso/head engagement with the bag and a front hard stop.

Output traces

The simulator returns time histories for displacement, acceleration, pressure, belt force, bag force, hard-stop force, chest deflection, HIC proxy, and deployment energy.

Simulator Information

Physics approximations

- 1D occupant excursion
- simplified crash pulse
- proxy chest and head injury measures

Metrics used

- mean HIC
- HIC CVaR90
- chest deflection
- HIC failure rate
- false deployment rate
- deployment energy

Failure-rate definition

$$\text{FailRate} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\text{HIC}_i > H_{\text{crit}}\} \times 100\%$$

where H_{crit} is the chosen HIC safety threshold used in evaluation.

All Algorithms Explained

Algorithm	Idea
Initial Design	Hand-chosen reference airbag algorithm.
Deterministic SQP	Single nominal crash case optimized with SLSQP.
Grid Search	Coarse discrete search over the algorithm space followed by local refinement on the best grid region.
MPC-Inspired Policy	Adaptive heuristic policy that changes deployment settings using scenario severity indicators.
Robust DE+SLSQP	Main stochastic optimizer using differential evolution for global exploration and SLSQP for local refinement.
Shift-Refined Local Search	Starts from the robust algorithm and performs fine local coordinate refinement on a larger shifted Monte Carlo sample.

Algorithm 1: Initial Design

Purpose

The initial design is the reference point used to measure improvement.

- It is a manually chosen fixed deployment profile.
- No optimization is performed for this algorithm.
- Algorithm values:
 - delay: 18 ms
 - pressure profile: 220 / 190 / 140 kPa
 - vent rate: 18
- How these values were chosen:
 - selected by engineering judgement as a feasible mid-range reference inside the allowed bounds
 - uses moderate trigger delay, high early pressure, then decreasing mid/late pressure to represent a conventional staged inflator profile
 - uses a moderate vent rate so the bag is neither too stiff nor too soft
- Mathematical view:

Algorithm 2: Deterministic SQP

Purpose

Deterministic SQP tests what happens if we optimize only for a single nominal crash case.

- Uses one representative crash scenario instead of a stochastic scenario set.
- Optimizer: SLSQP with bound constraints on delay, pressures, and vent rate.
- Mathematical view:

$$x_{\text{nom}} = \arg \min_{x \in \mathcal{X}} J(x; \omega_{\text{nom}})$$

where ω_{nom} is one nominal crash realization and SLSQP performs local constrained optimization.

- Advantage:
 - very fast
 - easy to interpret
- Limitation:
 - weak under uncertainty because it does not see rare or shifted crash scenarios

Algorithm 3: Grid Search

Purpose

Grid Search provides a strong non-gradient baseline over the fixed algorithm space.

- Searches a coarse grid over delay, pressure stages, and vent rate.
- Then performs a small refinement around the best coarse grid point.
- Mathematical view:

$$\mathcal{G} \subset \mathcal{X}, \quad x_{\text{grid}} = \arg \min_{x \in \mathcal{G}} \hat{J}(x)$$

followed by local refinement over a neighborhood $\mathcal{N}(x_{\text{grid}})$.

- Advantage:
 - robust to non-smooth simulation behavior
 - less likely to get stuck in one gradient basin
- Limitation:
 - expensive as dimensionality or grid resolution increases

Algorithm 4: MPC-Inspired Policy

Purpose

This algorithm is a simple adaptive baseline: instead of using one fixed airbag setting for every crash, it changes the settings based on crash severity.

- It looks at a few important crash features, such as vehicle speed, crash angle, seat position, belt status, and sensor delay.
- Using these inputs, it chooses a trigger delay, pressure profile, and vent rate.
- Mathematical view:

$$z(\omega) = [v, |\theta|, s_{\text{seat}}, \mathbf{1}_{\text{unbelted}}, \delta_{\text{sensor}}]$$
$$x(\omega) = \pi(z(\omega))$$

where π is a hand-designed function that converts crash features into deployment settings.

- Advantage:
 - adapts the airbag response to each scenario
- Limitation:
 - it is still heuristic, not a true MPC optimization

Algorithm 5: Robust DE+SLSQP

Purpose

This is the main stochastic robust optimizer.

- Stage 1: Differential Evolution explores the global algorithm space.
- Stage 2: SLSQP locally refines the best global candidate.
- Mathematical view:

$$x_{\text{rob}} = \arg \min_{x \in \mathcal{X}} [w_1 \mathbb{E}[\text{HIC}] + w_2 \mathbb{E}[D_{\text{chest}}] + w_3 \mathbb{E}[E_{\text{deploy}}] + w_4 \text{CVaR}_{90}(\text{HIC})]$$

with Monte Carlo estimation over training scenarios, DE for exploration, and SLSQP for local refinement.

- Advantage:
 - handles nonconvexity better than a single local optimizer
 - directly optimizes expected safety and HIC tail risk
- Limitation:
 - trained mostly on the default profile, so it may not be fully shift-aware

Algorithm 6: Shift-Refined Local Search

- Starts from the already good Robust DE+SLSQP solution.
- Samples a larger shifted Monte Carlo set.
- Performs coordinate-pattern local search with progressively smaller step sizes.
- Mathematical view:

$$x_{\text{ref}} = \arg \min_{x \in \mathcal{N}(x_{\text{rob}})} \frac{1}{|\mathcal{S}_{\text{shift}}^{\text{large}}|} \sum_{\omega \in \mathcal{S}_{\text{shift}}^{\text{large}}} J(x; \omega)$$

using coordinate updates

$$x^{k+1} = x^k \pm \alpha_k e_j$$

with shrinking step sizes α_k .

- Advantage:
 - makes microscopic algorithm adjustments near a strong robust solution
 - improved both iid and shifted results in the final run

Existing Algorithms and Novelty Opportunity

- We first implemented standard baselines: nominal local optimization, coarse global search, heuristic adaptive control, and robust stochastic optimization.
- These existing algorithms established two things:
 - nominal optimization alone was too weak under uncertainty
 - robust optimization helped a lot, but still left room under distribution shift
- That gap revealed the novelty opportunity: improve shift robustness without discarding the strong robust baseline.
- This led to one new algorithm:
 - **Shift-Refined Local Search**: starts from the robust solution and performs fine local search on a larger shifted sample
- So the novelty is not just adding another algorithm, but using the failure mode under shift to motivate targeted shifted-sample refinement.

Why the Final Algorithm Was Added

- The original robust optimizer already performed well under shift.
- A stronger algorithm was needed without overwriting the baseline robust result.
- The final added algorithm keeps the robust solution as a seed, then searches nearby points using a larger shifted scenario pool.
- This gives:
 - better shifted objective
 - lower shifted mean HIC
 - lower shifted failure rate
 - slightly improved iid results as well

Results: In-Distribution Evaluation

Algorithm	Objective	Mean HIC	CVaR90	Fail Rate
Initial Design	0.431	259.89	811.40	16.00%
Deterministic SQP	0.431	259.85	811.40	16.00%
Grid Search	0.379	249.09	752.15	14.33%
MPC-Inspired Policy	0.415	247.59	751.79	12.67%
Robust DE+SLSQP	0.356	230.41	706.29	11.67%
Shift-Refined Local Search	0.353	226.52	691.21	11.00%

Definition of fail rate

Fail rate is the percentage of evaluation scenarios whose HIC exceeds the chosen safety threshold.

Best iid algorithm

Shift-Refined Local Search achieved the best objective and the lowest failure rate on the default distribution.

Results: Shifted Evaluation

Algorithm	Objective	Mean HIC	CVaR90	Fail Rate
Initial Design	0.892	444.16	1312.55	32.00%
Deterministic SQP	0.891	443.96	1311.02	32.00%
Grid Search	0.738	401.87	1153.82	28.67%
MPC-Inspired Policy	0.767	412.99	1189.62	27.33%
Robust DE+SLSQP	0.589	377.17	1144.52	21.00%
Shift-Refined Local Search	0.579	372.21	1142.55	20.33%

Best shifted algorithm

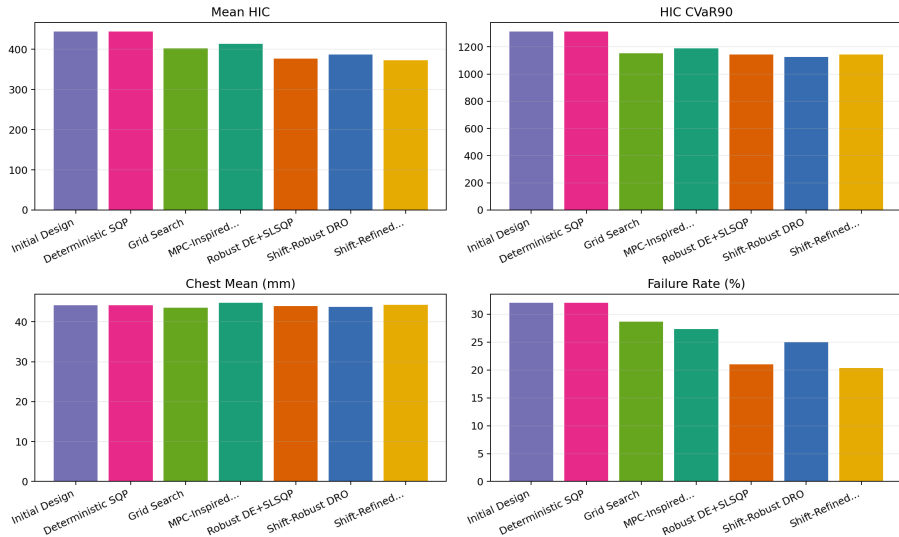
Shift-Refined Local Search gave the best shifted objective and improved over Robust DE+SLSQP on mean HIC and failure rate.

Best Algorithms from the Final Run

Algorithm	Delay	Early	Mid	Late	Vent
Initial Design	18.00	220.00	190.00	140.00	18.00
Grid Search	16.00	300.00	300.00	100.00	4.00
Robust DE+SLSQP	25.89	299.16	286.95	102.73	4.41
Shift-Refined Local Search	25.39	300.00	300.00	120.73	4.00

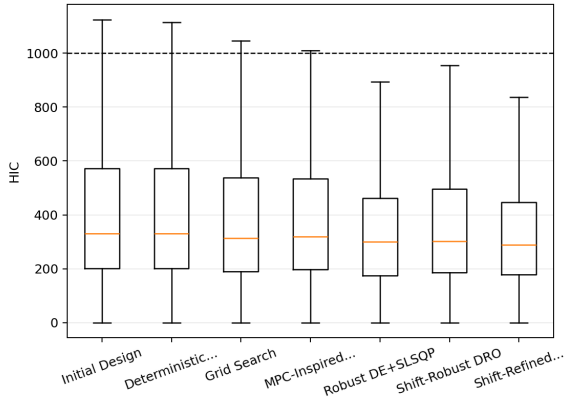
- Stronger algorithms consistently prefer high early and mid pressures.
- Best fixed algorithms also push the vent rate close to its lower bound.

Comparison Plot: Algorithm Metrics

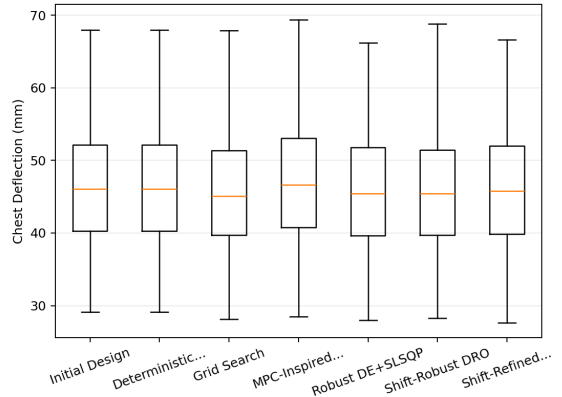


Comparison Plot: Distribution Spread

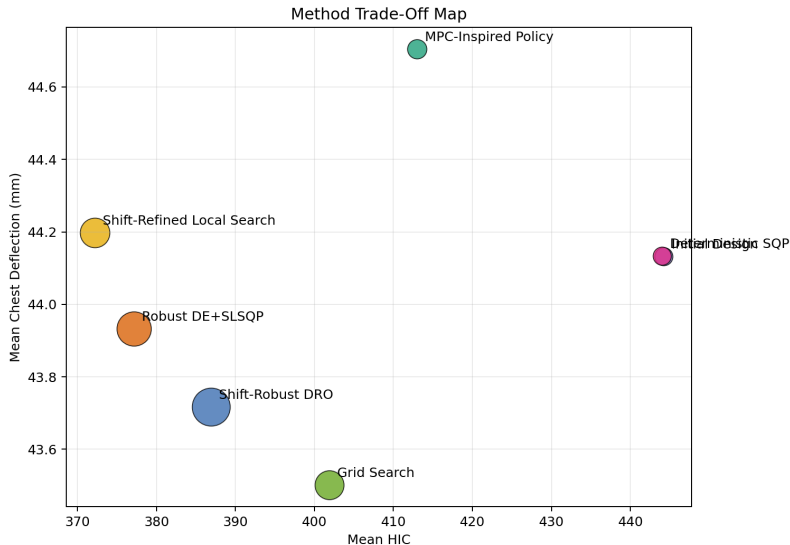
Scenario HIC Spread



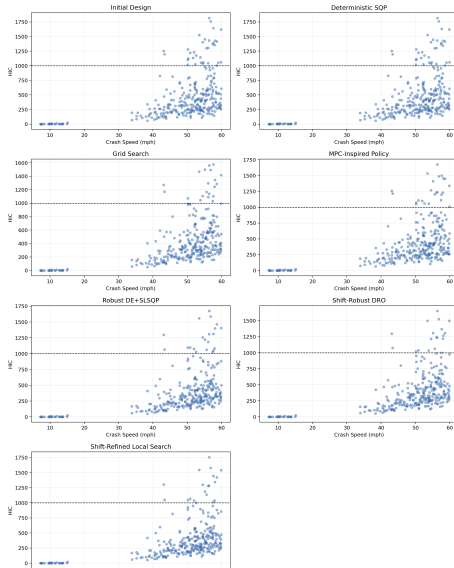
Scenario Chest Deflection Spread



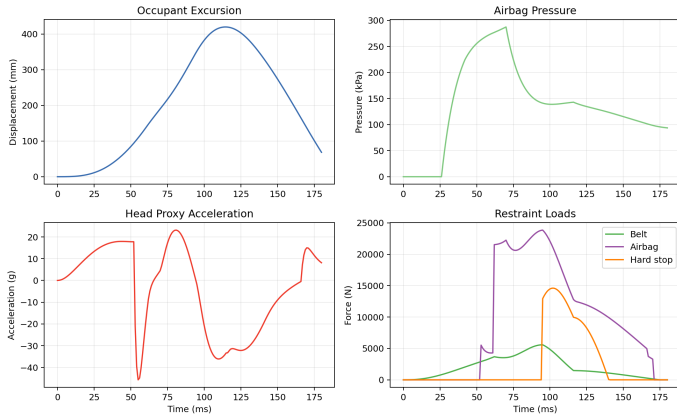
Comparison Plot: Trade-Off Map



Comparison Plot: Speed vs HIC



Representative Simulation Trace



Key Observations

- 1 Deterministic SQP is almost identical to the initial design, so nominal optimization alone is not enough.
- 2 Grid Search is a strong baseline, showing that simple search can already find good high-pressure / low-vent algorithms.
- 3 Robust DE+SLSQP gives a large improvement over the initial design under both iid and shifted testing.
- 4 The final shift-refined local-search algorithm gave the best overall performance, especially on shifted evaluation.

Conclusions

- We built an end-to-end stochastic airbag optimization framework from the project brief.
- The framework includes scenario generation, reduced-order crash simulation, multiple optimization baselines, and automated reporting.
- Distribution shift matters: algorithms must be tested beyond the default training distribution.
- Robust DE+SLSQP is a strong baseline, but local refinement on a larger shifted sample improved it further.
- Final best algorithm: **Shift-Refined Local Search**.

- Improve physical fidelity beyond the reduced-order model.
- Perform ablation studies on shifted-sample size and objective weights.
- Develop a stronger adaptive controller than the current MPC-inspired heuristic.
- Extend the analysis to richer occupant classes and crash geometries.

Thank You

How Each Objective Term Is Calculated

- **Mean HIC, $\mathbb{E}[\text{HIC}]$:**

$$\text{HIC} = \max_{t_2 - t_1 \leq 36 \text{ ms}} \left[(t_2 - t_1) \left(\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} a(t) dt \right)^{2.5} \right]$$

$$\mathbb{E}[\text{HIC}] \approx \frac{1}{N} \sum_{i=1}^N \text{HIC}_i$$

- **Mean chest deflection, $\mathbb{E}[D_{\text{chest}}]$:**

- Belt force, torso airbag force, and hard-stop force drive a chest-deflection state.

$$D_i = \max_t d_i(t) \quad \mathbb{E}[D_{\text{chest}}] \approx \frac{1}{N} \sum_{i=1}^N D_i$$

where $d_i(t)$ is the chest-deflection trace for scenario i .

How Each Objective Term Is Calculated

- **Mean deployment energy, $\mathbb{E}[E_{\text{deploy}}]$:**

- At each timestep, the model estimates inflator energy from pressure rise and vent loss.

$$E_i = \int_0^T \dot{E}_i(t) dt \quad \mathbb{E}[E_{\text{deploy}}] \approx \frac{1}{N} \sum_{i=1}^N E_i$$

- **Tail risk, $\text{CVaR}_{90}(\text{HIC})$:**

$$q_{0.9} = \text{Quantile}_{0.9}(\text{HIC})$$

$$\text{CVaR}_{90}(\text{HIC}) = \mathbb{E}[\text{HIC} \mid \text{HIC} \geq q_{0.9}]$$